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Human identification system using 3D skeleton-based gait features and LSTM model☆

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ABSTRACT

Vision-based gait emerged as the preferred biometric in smart surveillance systems due to its unobtrusive nature. Recent advancements in low-cost depth sensors resulted in numerous 3D skeleton-based gait analysis techniques. For spatial–temporal analysis, existing state-of-the-art algorithms use frame-level information as the timestamp. This paper proposes gait event-level spatial–temporal features and LSTM-based deep learning model that treats each gait event as a timestamp to identify individuals from walking patterns observed in single and multi-view scenarios. On four publicly available datasets, the proposed system stands superior to state-of-the-art approaches utilizing a variety of conventional benchmark protocols. The proposed system achieved a recognition rate of greater than 99% in low-level ranks during the CMC test, making it suitable for practical applications. The statistical study of gait event-level features demonstrated retrieved features' discriminating capacity in classification. Additionally, the ANOVA test performed on findings from K folds demonstrated the proposed system's significance in human identification.

1. Introduction

In smart environments (smart building, smart campus, smart city, etc.), the need for reliable, non-contact human identification at a distance is dramatically increasing in various applications embedded in intelligent surveillance systems. These systems play a major role in many practical applications like forensics, terrorist monitoring, crime prevention, and access control [1]. Identifying a known person or person of interest from a group of people using their biometric traits is of utmost importance in these applications. Most biometrics, including the face, iris, and the fingerprint, require high-quality images or physical contact to obtain reliable data [2].

Gait is a behavioral biometric that has received more attention recently for human identification. Boyd et al. [3] defined gait as a coordinated, cyclic combination of movements that result in human locomotion. Kastaniotis et al. [4] explained that the entire gait cycle has two phases, namely: Stance and Swing, considering any leg as a reference point. 60% of the gait cycle is composed of Stance, and the remaining 40% is the Swing phase. These phases are further divided into several events. The gait cycle begins with the Initial Contact (IC) event in the Stance phase and ends with the Swing phase's Terminal Swing event. The various events and their proportion in a gait cycle are shown in Fig. 1. We adapted the timing distribution of various events in the gait cycle from [5]. Research on gait has drawn the attention of several researchers due to the following key benefits [6]:

- Each individual has a distinct walking style that is difficult to conceal or imitate.
- Its unobtrusive nature, i.e., gait data are collected at a distance without the subject's knowledge.
- It is much more difficult to continuously alter a person's gait characteristics.

Most of the approaches proposed for gait recognition over the years are video-based. These video-based approaches fall into two categories, namely: model-free and model-based [7].

Model-free methods focus on extracting static and dynamic features directly from images of the walking sequence. The majority of these methods obtain gait information by utilizing silhouette image-based features of the human body. Han and Bhanu [8] proposed a spatial–temporal representation of the gait cycle called Gait Energy Image (GEI) for individual recognition. Liu and Zheng [9], proposed Gait History Image (GHI) to represent more insightful gait motion by preserving the temporal information associated with human motion. However, silhouette quality, i.e., segmentation errors, affect the performance of GEI [10]. Deng et al. [11] proposed a gait recognition algorithm based

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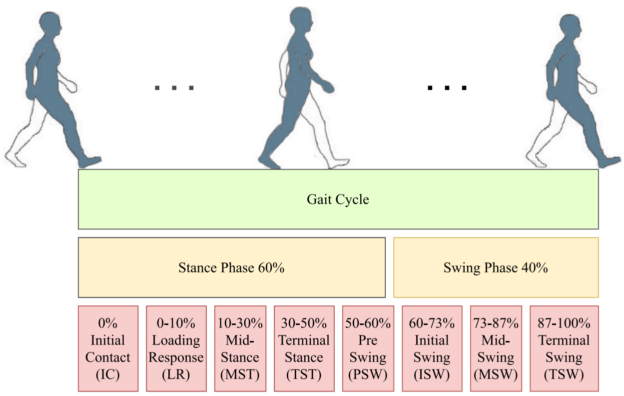


Fig. 1. Normal gait cycle phases, events, and their timings.

on the combination of spatial–temporal and kinematic gait features extracted from silhouette images.

Model-based approaches analyze the shape and kinematics of human body parts for gait recognition. The majority of these model-based gait recognition methods focused on distances or joint angles on the human body for feature extraction [12]. The recent advancements in 3D visual depth sensors, such as Microsoft Kinect [13], has resulted in a dramatic increase in the number of 3D model-based approaches being introduced. Kinect depth sensors can capture data in multiple modalities. 3D skeleton joint data provided by Kinect depth sensor eliminates the need for complex procedures of building a model from visual data streams [14]. Numerous skeleton-based gait recognition systems have been proposed in the recent years [2,6,14–16]. Additionally, the success of deep learning [17] models on image/video-based classification tasks prompted researchers to conduct new investigations on video-based human identification systems using deep learning.

Despite the abundance of research on gait-based human identification, there is an inadequate exploration of the gait event-level features for human identification from freestyle walking. Also, gait recognition is still one of the challenging tasks as it needs to handle the occlusion, the difference in clothing, and temporal aspects of various features during motion, etc. This motivated us to propose an efficient scheme to cope with noisy and occluded joints and a deep learning model based on LSTM that learns from gait events for gait cycle classification, followed by fusion to identify the person. For this paper, the main contributions are as follows:

- Represented each gait event as a timestamp in the entire gait cycle. Also, proposed a set of gait event-level features to capture the spatial–temporal information of the gait cycle.
- Formed a quantitative summary of various features extracted from frames in each timestamp to minimize the effects of noise and the occlusions. As a result, the proposed human identification system's total computational cost is significantly reduced.
- Proposed a deep learning model based on LSTM [18] to efficiently recognize the gait cycle using features extracted from the sequence of gait events.

The remaining part of the paper is organized as follows. Section 2 reviews the study on related work on gait recognition with a particular emphasis on skeleton-based techniques. The proposed system for human identification is discussed in Section 3. Section 4 discusses the benchmark datasets used to evaluate the model performance. Experimental findings and analysis are presented in Section 5. Finally, Section 6 concludes the paper with a discussion of future directions.

2. Related works

In the era of smart environments, identifying a person automatically from a group of people is one of the most important tasks for ensuring a secure and comfortable life. Vision-based gait recognition identifies a person by analyzing visual data of a pedestrian walking acquired by a camera. Due to page constraints, we confine our discussion to existing works on human identification using 3D skeleton-based gait features.

Numerous studies have established the importance of 3D skeleton-based gait features in human identification. Several early approaches introduced in the literature, such as Araujo et al. [20], demonstrated promising results using only static data. The authors developed a system for human identification using relative lengths of several body parts extracted from 3D skeleton data and a K-Nearest Neighbor (KNN) classifier. Dikovski et al. [21] suggested a method for combining static and dynamic features to improve the performance. Anthropometric data along with gait data were used in [10,22] for better recognition accuracy. Additionally, Andersson and Araujo [22] created a dataset of 164 subjects. Yang et al. [10] derived a set of features using relative distances between the joints. Finally, the combined distance and anthropometric features are classified using the KNN classifier, and further, a majority voting fusion scheme is applied. Though the method considered gait cycles for classification, it used all the features extracted from each frame in the gait cycle. Kastaniotis et al. [23] proposed a hierarchical representation of gait trajectories for human identification by adapting an action recognition framework to capture dynamic gait features effectively. Though these approaches demonstrated the direction of utilizing 3D skeleton data for human identification, they require further improvement to account for occlusion, noise, and other factors. Kastaniotis et al. [4] proposed a pose-based representation by combining Euclidian and Riemannian features. Also, this work created a publicly available dataset with 30 participants walking in a straight line. Deng and Wang [14] employed a deterministic learning theory to capture the gait dynamics and performed feature fusion for human identification. However, these approaches did not account for the differences in feature distributions associated with various gait events. Khamsemanan et al. [6] used posture-based features and classification to identify a person from a free walking sequence. For classification, a feature vector is constructed from the displacements of all joints in adjacent frames and the center-of-body relative joint coordinates. Later, a fusion of the predictions from each frame is considered for final judgment. Because features are collected from individual frames, this results in a greater number of features being processed and affected by occlusion and noise. Choi et al. [2] proposed an approach to minimize the effect of noise and frame-level feature mapping to find the similarities between the gait cycles. This approach, however, is limited by the absence of dynamic gait characteristics.

Various deep learning-based models with extracted features were proposed in the literature. Bari and Gavrilova [15] introduced joint relative triangular area and joint relative cosine dissimilarities to capture the pose and view-invariant features. Also, they proposed a deep learning model to classify the multiple gait cycles extracted from the walking sequence. However, the deep learning model introduced by these authors has a large number of trainable parameters. Hosni and Amor [24] suggested a geometric convolutional encoder–decoder network. An LSTM based neural network to learn skeleton information from each frame in the sequence is proposed in Li et al. [25]. This method used raw data without considering gait-specific features. Liu et al. [19] suggested a CNN-LSTM based deep learning model that captures spatial and temporal information and is followed by a Support Vector Machine (SVM) classification on combined features. Limcharoen et al. [26] removed noise by averaging co-ordinate points from successive frames and proposed a LSTM model for gait recognition. Further, several recent works on image/video data for better feature extraction and representation are found in [27–30]. Table 1 summarizes the key features of existing works that motivated us to propose this work.

From the literature survey, it is clear that many researchers introduced various approaches for gait-based human identification. Visual data is now captured in various modalities, including the 3D skeleton, depth, and RGB, resulting from advancements in visual sensing devices.

Table 1
Key features of existing works on skeleton data based gait analysis.

Methods	Feature Extraction	Classification Method	Remarks
Khamseman an et al. [6]	Posture-based feature vectors are generated for each frame.	Supervised learning algorithms for classifying each frame followed by score fusion.	Extracted 80 features from each frame of the walking sequence. The features are well suited for handling the observation angle issue. The feature vectors from noisy frames will affect the performance of the human identification system.
Bari and Gavrilova [15]	Feature vectors are generated for each gait cycle from frames within the gait cycle.	Deep learning Model with Dense, Batch Normalization, and Dropout.	Performed individual gait cycle recognition. The deep learning model has more parameters will lead to higher computational costs.
Limcharoen et al. [16]	Feature vectors contain the joint replacement coordinates using a set of selected frames.	CNN-based deep learning model.	Focused on multi-viewpoint challenges in gait recognition for view-independent gait recognition. Further, a few local joint movements were considered. To improve the performance, there is a need to consider the entire body movement information.
Liu et al. [19]	Combination of skeleton gait energy image and dynamic features from all frames.	Two separate deep learning models based on CNN and LSTM, followed by fusion.	Captured both spatial and temporal information using two separate deep learning models but this will lead to the higher computational cost of the overall system.
Proposed System	Feature vectors are generated for each gait cycle event.	LSTM based deep learning model.	A cost-effective and an efficient approach with few features and a deep learning model using the LSTM with few parameters for gait cycle recognition and the human identification from the entire walking sequence based on the score fusion. Thus the proposed system is not only efficient in human identification but also reduces the computational cost by 24 times as compared to the existing work of [15].

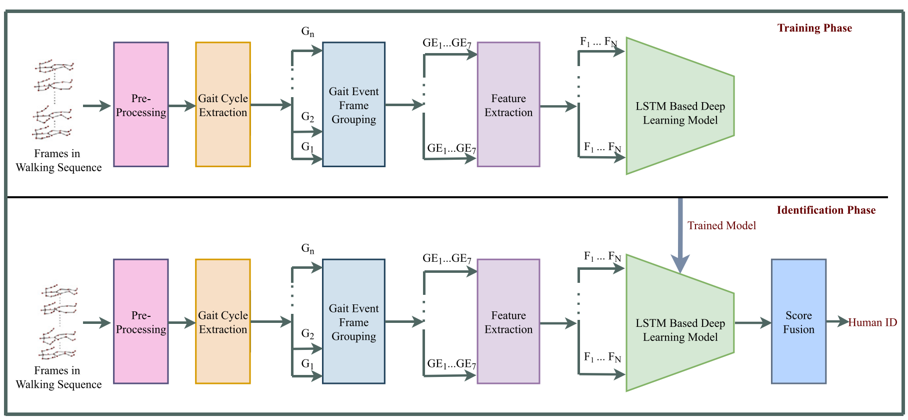


Fig. 2. The workflow of the proposed human identification system using 3D skeleton-based gait features and the LSTM model.

In comparison to RGB videos, skeleton-based data is superior in terms of background adaptability, resistance to light variations, noise, and minimal computational cost, [31,32]. Due to this, nowadays, the 3D skeleton-based gait analysis is sought-after research. Few approaches were proposed to capture temporal information in the gait data; none of the methods considered the gait event from the temporal aspect. Some of the existing works used frame-level prediction fusion for final prediction, leading to more computation and being affected by several noisy and occluded frames in the walking sequence. This motivated us to propose an efficient scheme to cope with noisy and occluded joints and a deep learning model based on LSTM that learns from gait events for gait cycle classification, followed by fusion to identify the person.

3. Proposed system

The proposed system's primary objective is to identify a person based on their normal walking sequence. Fig. 2 shows the overall workflow of the proposed system. The workflow is divided into two phases, namely: the training phase and the identification phase. The training phase generates a deep learning model to recognize the gait cycles. The identification phase obtains the matching probability score by utilizing the deep learning model constructed during the training phase. Both training and identification phases have a common set of operations in the first few subsystems. Each phase begins with the extraction of gait cycles from a person's walking sequence, followed by the extraction of features from individual gait events to generate the final feature vector representing the complete gait cycle. Further, the proposed LSTM based deep learning model is trained using the extracted features to recognize the gait cycles. Finally, in the identification phase, the probability scores of gait cycle recognition obtained from the trained LSTM model are combined using a score fusion technique to identify the exact person. The remainder of this section describes each subsystem in detail.

3.1. Pre-processing

The distance between person and camera varies in a scenario with a still camera capturing the walking sequence. The Kinect depth sensor provides 3D skeleton data in different scales depending on the subjects' placement in the data collection environment. So initially, some measures are taken to align the skeletons in a new global coordinate system, and the details are as follows.

3.1.1. Align skeleton data in new global coordinate system

Consider a sequence of skeleton frames, $F = \{f_1, f_2, \ldots, f_n\}$ where, n is the number of frames captured by the Kinect depth sensor from the entire walking sequence. Each frame gives information about a set of P 3D joint coordinates of the skeleton, $J = \{j_1, j_2 \ldots, j_P\}$. Each joint contains information about three coordinates (x, y, z). Fig. 3 shows the skeleton format captured by the Kinect depth sensor V1, along

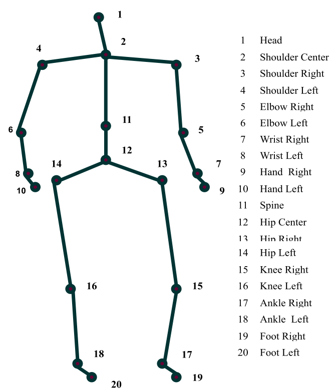


Fig. 3. Skeleton joints extracted by Kinect depth sensor V1.

with the names of each joint. As the person moves, the values of these coordinates will vary. Initially, a new global coordinate system is constructed and made hip-center as the center of the coordinate system and is fixed at $(0, 0, 0)$. Further, all other joints are translated to the new global coordinate system using the following steps.

Let (h_x, h_y, h_z) be the hip center coordinate values along x, y, and z axes, respectively, in a skeleton frame, and represented as a column vector: $\begin{bmatrix} h_x \\ h_y \\ h_z \end{bmatrix}$. To translate this with the origin $\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$ of the global coordinate system, the translation distances are defined as $t_x = -h_x$, $t_y = -h_y$ and $t_z = -h_z$. Then all joints j_i for all $i = 1$ to P, in the original 3D coordinate system, are transformed to the new global coordinate system using the transformation matrix as shown in Eq. (1).

$$\begin{bmatrix} x'_i \\ y'_i \\ z'_i \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & t_x \\ 0 & 1 & 0 & t_y \\ 0 & 0 & 1 & t_z \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_i \\ y_i \\ z_i \\ 1 \end{bmatrix}, \tag{1}$$

Where, x_i, y_i, z_i are the 3D coordinate values of i^{th} joint in original coordinate system, and x'_i, y'_i, z'_i are the corresponding values in the new global coordinate system.

3.1.2. Data normalization

Following translation, to cut down the effect of the coordinate difference in the skeleton data, we normalized each skeleton joint from the ith walking sequence, kth frame number, and jth joint number $C_{ikj} = (x, y, z)$ concerning the entire dataset as specified in Eq. (2).

$$C'_{ikj} = \frac{C_{ikj} - c_{min}}{c_{max} - c_{min}} * 10, \tag{2}$$

Where, C'_{ikj}, c_{min}, and c_{max} are the normalized skeleton joint coordinate, minimum coordinate value, and the maximum coordinate value, respectively. The dataset dependent parameters c_{min}, and c_{max} are determined using the entire dataset.

3.2. Gait cycle extraction

Walking is a series of repetitive limb motions that assist the body in moving forward. In other words, it is a series of gait cycles, each beginning with a heel strike (Initial Contact) of one leg to the floor and ending with another heel strike (Terminal Swing) of the same leg. The first 60% of the gait cycle is the Stance phase, while the remaining 40% is the Swing phase. Further, researchers have identified seven different events in the gait cycle, as shown in Fig. 1. In each of these events, limbs are positioned in specific ways to exhibit characteristics that aid in gait recognition.

The left and right ankles remain at their maximum distance during heel strikes. In this study, we determined the gait cycle using ankle distance. Initially, the Euclidean distance between the left and right ankles in each frame in the walking sequence is computed by using Eq. (3). Where, (x_l, y_l, z_l) and (x_r, y_r, z_r) represent the (x, y, z) coordinates of left and right ankle joints, respectively. Three consecutive peaks in the ankle distance-vector define a gait cycle.

$$D = \sqrt{(x_l - x_r)^2 + (y_l - y_r)^2 + (z_l - z_r)^2}, \tag{3}$$

The noise in the distance vector is minimized using Savgol filter [33] with a window size of 11. Fig. 4(a) shows the original and smoothed graph of the ankle distance-vector generated from a sample walking sequence taken from the Kinect Gait Biometry Dataset (KGBD) [22]. The corresponding smoothed ankle distance-vector peaks are shown in Fig. 4(b). The local peaks with a height less than $0.4 \times$ maximum peak height in the sequence are discarded. The proposed system extracted all possible gait cycles and filtered out the frames that do not contribute to a normal gait cycle.

3.3. Gait event frame grouping

Gait features are unique to each person. Human gait shows a distinct movement of limbs during different periods of the gait cycle. Authors in [2], and [6] proved that each frame in the gait cycle contributes to the gait feature vector. Each event in the gait cycle is distinct in its own way. Thus, the proposed human identification system treats gait events as timestamps and extracts features from each gait event. Fig. 1 depicts the timing distribution of each gait event.

Initially, a set of frames forming a gait cycle are distributed among seven sets based on the gait event period. Later from each group, feature vectors of the same size are extracted. The proposed work computes various sets of inter and intra-frame joint distance and angle features from each group. To minimize the effect of occluded joints in the frame, we utilized the average and standard deviation of corresponding features collected within the frame group instead of using the raw features collected from the frames. The following subsection elaborates more on the extracted features.

3.4. Feature extraction

During walking, each person exhibits unique upper and lower limb motions. The distance and angle between the 3D skeleton joint coordinates vary as limbs move. The proposed work extracts a set of inter and intra-frame Euclidean distance and angle between 3D joint coordinates. The Euclidean distance between any two 3D joint coordinates $A = (x_1, y_1, z_1)$ and $B = (x_2, y_2, z_2)$ is calculated using Eq. (4).

$$D = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2 + (z_1 - z_2)^2}, \tag{4}$$

Given three joint coordinates $A = (x_1, y_1, z_1)$, $B = (x_2, y_2, z_2)$, and $C = (x_3, y_3, z_3)$ in a 3D coordinate system, then the angle $\Theta = \angle ABC$ is obtained by using Eq. (5).

$$\Theta = cos^{-1}\left(\frac{\vec{AB}.\vec{CB}}{\|\vec{AB}\|\|\vec{CB}\|}\right), \tag{5}$$

3.4.1. Body Part Length Features (BpLF)

We considered seven body part lengths as features in the proposed work. The various body part lengths considered are depicted in Fig. 5(a). The variation in these lengths must be very small within the sequence of frames. So, at first, for each frame f_i in a gait event, the body part length is computed, and then, the mean of each of these body part lengths from frames in gait event are considered as features.

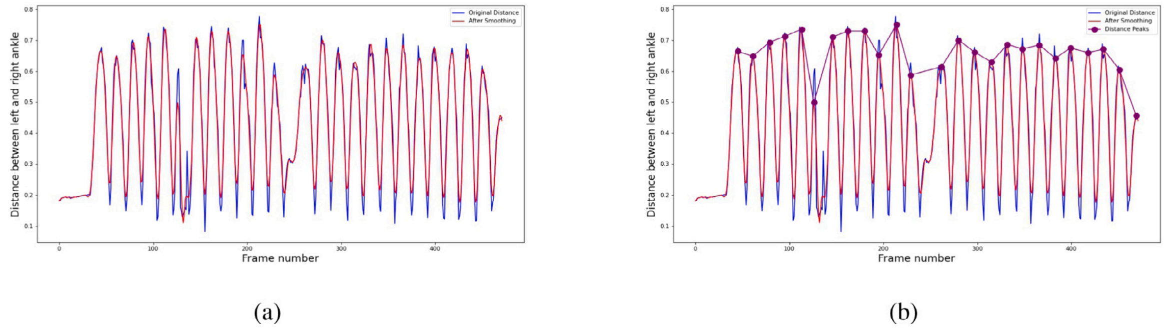


Fig. 4. Gait cycle extraction steps: (a) Original and smoothed ankle distance-vector (b) Peaks in smoothed ankle distance-vector.

3.4.2. Joint Distance Features (JDF)

The relative distance between the body joints changes significantly as the person walks. So the mean and standard deviation of body joint distance s from frames in a gait event is computed. The joint distances used to feature construction are shown in Fig. 5(b). Collectively we extracted 18 mean and 18 standard deviation features.

3.4.3. Joint Angle Features (JAF)

To focus on the body joint angle during walking, in the proposed work, we concentrate on the 15 joint angles depicted in Fig. 5(c). The mean and standard deviation of the joint angle i from frames in gait events are computed. This contributes 30 features to the feature set. The joint angle between three joints is computed as given in Eq. (5).

3.4.4. Inter-Frame Joint Distance Features (InJDF)

Inter-frame joint distance features are proposed to keep track of the relative change in the position of a body joint between successive frames. Each inter-frame joint distance is computed using Eq. (6). Where, InD is the inter-frame joint distance of skeleton joint i between kth and $(k+1)$th frames in a gait event, and $k = \{1, 2, \dots, n-1\}$, where n is number of frames in gait event. Finally, we compute the mean and standard deviation of these distances for each joint i taken into account for each gait event. Fig. 5(d) shows the 16 inter-frame joint distances considered in this work. This contributes 32 features to the feature vector.

$$InD = \sqrt{(x_i^k - x_i^{k+1})^2 + (y_i^k - y_i^{k+1})^2 + (z_i^k - z_i^{k+1})^2}, \tag{6}$$

3.4.5. Inter-Frame Joint Angle Features (InJAF)

Additionally, we propose inter-frame joint angle features for tracking the difference in joint angle between successive frames. First, the joint angles depicted in Fig. 5(c) are computed in every frame. Later, the difference between these angles is computed for each successive frame. The overall procedure is given in Eq. (7). Where, InA_B is the difference between $\angle ABC$ in kth and $(k+1)$th frames in a gait event, A, B, C are skeleton joints and $k = \{1, 2, \dots, n-1\}$, where n is number of frames in gait event. Finally, we determine the mean and standard deviation of each of these angle differences from the gait event. This contributes 30 features to the feature vector.

$$InA_B = cos^{-1}\left(\frac{A^{k+1}\vec{B}^{k+1}.C^{k+1}\vec{B}^{k+1}}{\|A^{k+1}\vec{B}^{k+1}\|\|C^{k+1}\vec{B}^{k+1}\|}\right) - cos^{-1}\left(\frac{A^k\vec{B}^k.C^k\vec{B}^k}{\|A^k\vec{B}^k\|\|C^k\vec{B}^k\|}\right), \tag{7}$$

3.5. LSTM based deep learning model

The gait cycle comprises a repeated sequence of gait events. To effectively use the temporal features contained within a gait cycle, we proposed a deep learning model composed of two layers of LSTM. Fig. 6 illustrates the proposed deep learning model. The proposed LSTM based deep learning model is implemented in Keras [34] as a sequential

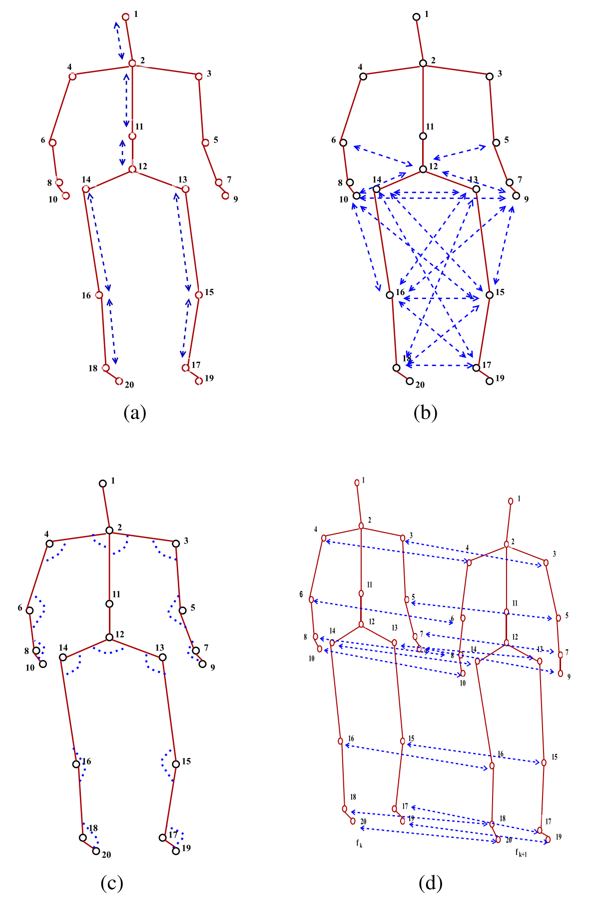


Fig. 5. Extracted Features: (a) BpLF (b) JDF (c) JAF (d) InJDF.

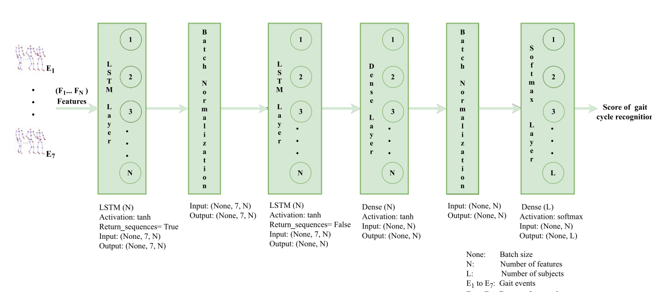


Fig. 6. Proposed LSTM based deep learning model.

model. The final model comprises two LSTM layers, a Dense layer with N units and *tanh* activation function, two Batch Normalization layers, and a Softmax layer. The input data is divided into seven timestamps since we considered seven events from each gait cycle. The human identification labels are also one-hot encoded and fed to the neural network along with features. Before providing the features to the first LSTM layer, we reshaped the features from gait cycles into $(7, N)$, where N is the number of features from each gait cycle event. The first LSTM layer's output is normalized using the first Batch Normalization layer to reduce overfitting, thereby getting the more stable model. The second LSTM layer is fed the normalized output from the first Batch Normalization layer. Further, to boost the recognition performance, the Dense layer processes the second LSTM layer's output. The Dense layer's output is normalized using the second Batch Normalization layer before the softmax layer to accelerate the training process. Using the hyperbolic tangent function, we introduced non-linearity into the neural network model. Finally, the Softmax layer gives the probability score of classification. The optimal weights are obtained by using the Adam optimizer [35].

3.6. Score fusion

For each gait cycle, the proposed deep learning model's Softmax layer generates the probability score vector $P = \{p_1, p_2, \dots, p_L\}$ for identifying the human. Then, using Eq. (8), the final prediction of an identified person (human) is performed via score fusion as shown in Eq. (8).

$$P_{id} = argmax(\sum_{1}^{g} p_1, \sum_{1}^{g} p_2 \ldots, \sum_{1}^{g} p_L), \tag{8}$$

Where, g, L, p_i, and P_{id} are the number of gait cycles extracted from the walking sequence, the number of subjects in the dataset, a score of identification of ith person, and the person identified from the walking sequence, respectively. The function *argmax* gives the position of the maximum argument. The overall algorithm (Algorithm 1) is given below:

Algorithm 1: Human identification from skeleton data based gait dataset

```
function Human_identification (W);
Input : A set of walking sequences of people W
Output: Person identified P_id
foreach walking sequence w ∈ W do
    Extract sequence of skeleton frames F
    N= Number of frames in F
    Align skeleton data from each frame in a global coordinate system using
      Equation (1).
    Normalize the skeleton data from each frame using Equation (2).
    Extract a set of gait cycles G from F based on distance between ankles as
      given in Section 3.2.
    foreach gait cycle g ∈ G do
        Group frames in g under different gait events:
          E = {LR, MST, TST, PSW, ISW, MSW, TSW} based on proportion
          of frame distribution as mentioned in Fig. 1.
        Assign Gait cycle feature vector GFE = []
        foreach gait event e ∈ E do
            Assign gait event feature vector FE = []
            //Perform feature extraction
            Extract BpLF, JDF, JAF, InJDF, and InJAF features using the
              procedure given in Section 3.4
            FE = [BpLF, JDF, JAF, InJDF, InJAF]
            GFE = [GFE, FE]
        end
    end
end
Partition the feature vectors into train and test sets based on specifications of
  Evaluation Protocol.
Train the proposed LSTM based deep learning model.
Feed feature vector of test set to fitted proposed LSTM based deep learning model.
if Score fusion required based on Evaluation Protocol then
    Perform Score fusion as given in Equation (8).
    Return Person Identified P_id.
end
else
    foreach gait cycle g ∈ test do
        Return Person identification label which obtained highest score.
    end
end
```

4. Datasets

We evaluated the proposed system's performance against four publicly available benchmark 3D skeleton-based gait datasets, namely: Kinect Gait Biometry Dataset (KGBD) [22], UPCV1 [23,36], UPCV2 [4], and KS20 VisLab Multi-View Kinect skeleton dataset [37,38].

4.1. Kinect Gait Biometry Dataset (KGBD)

The KGBD is a collection of 164 subjects' 3D skeleton-based gait data. Kinect was positioned in the center of a semi-circle with a spinning dish to track the subject's movement and direct the subject to walk in a semi-circular path during data collection. The majority of the walking sequences range from 500 to 650 frames. Except for a few, each participant has five walking sequences. Each frame contains 3D coordinate data of 20 joints. We extracted 8–29 gait cycles from each walking sequence.

4.2. UPCV1

The UPCV1 gait dataset is captured using Kinect V1 depth sensor. It contains five walking sequences from each of 15 male and 15 female subjects. Each person walks in a straight direction and has 50–120 frames per walking sequence. Most walking sequences have two gait cycles.

4.3. UPCV2

The UPCV2 gait dataset was collected using a Microsoft Kinect V2 depth sensor, with each skeleton having 25 joints. It contains ten walking sequence recordings of 30 individuals (17 males and 13 females) walking in a straight line. Most of the walking sequences contain three gait cycles.

4.4. KS20 VisLab multi-view kinect skeleton dataset

A dataset of multiple views of gait was captured with the Microsoft Kinect V2 depth sensor. It consists of gait cycles of 20 subjects captured from five different views (left lateral at 0°, left diagonal at 30°, frontal at 90°, right diagonal at 130°, and right lateral at 180°). Three samples are taken from each viewpoint, per subject, with each sample containing a single gait cycle.

5. Experimental results and analysis

The LSTM model is trained with different feature combinations during the training phase to study various features' contributions to gait recognition. The fitted LSTM model is used during the identification phase. The proposed score fusion is performed on the output from the softmax layer based on its suitability for evaluation protocols. Score fusion is skipped in some protocols where it is not playing a role. On these datasets, a series of experiments are conducted using various classification algorithms, including KNN, SVM, Random Forest (RF), two-layer Artificial Neural Networks (ANN), and a variant of LSTM called Gated Recurrent Units (GRU) [39].

Table 2
Rank-1 recognition performance of the proposed system for human identification with different feature combinations using $Protocol - 1$ (in %) (Best results are in bold).

Experimental Setup	Gait Dataset											
	KGBD				UPCV1				UPCV2			
	Accuracy	Precision	Recall	F1-Score	Accuracy	Precision	Recall	F1-Score	Accuracy	Precision	Recall	F1-Score
InJDF (30 features) + LSTM	72.9	63.2	73	65.8	59.99	49	60	52.4	80.32	72.3	80.5	74.7
InJDF + InJAF (62 features) + LSTM	91.32	87.4	91	88.4	59.99	48.8	60	52.4	88.32	83.3	88.5	84.8
BpLF+JDF+ JAF (73 features) + LSTM	97.56	96.6	97.6	96.66	95.9	94	96	94.8	98.33	97.5	98.4	97.9
BpLF+JDF+ InJDF+JAF (105 features)) + GRU	98.53	97.8	98.6	98	95.99	94	96	94.8	97.33	96	97.4	96.6
BpLF+JDF+ InJDF+JAF (105 features)) + LSTM (Proposed System)	**98.54**	**97.6**	**98.6**	**97.8**	**98.00**	**97**	**98**	**97.4**	**98.33**	**97.5**	**98.5**	**98**

5.1. Experiments on single-view dataset

The proposed system is evaluated on three single-view datasets based on the following two existing protocols:

- $Protocol - 1$: Human identification using the walking sequence based on the approach of [6].
- $Protocol - 2$: Gait cycle recognition based on the approach of [15].

In both protocols, the proposed system's performance was evaluated using K-fold cross-validation using the metrics, namely: Accuracy, Precision, Recall, and F1-Score. Precision is the measure of positive class predictions that are actually in a positive class. The Recall is the positive class sample retrieved among all positive class samples. The F1-Score is a model performance metric that combines Precision and Recall. The Precision, Recall, and F1-Scores for each class are initially computed, and the corresponding weighted averages for each fold were determined. Finally, the averages of these metrics on K folds are reported in the paper.

5.1.1. Protocol − 1

Our proposed system's ultimate goal is to identify a person using the walking sequence. We used the entire walking sequence for human identification based on the work of [6]. The dataset is divided into K disjoint subsets or folds based on the number of walking sequences for each person. So we used 5-fold cross-validation on KGBD and UPCV1 datasets and 10-fold cross-validation on the UPCV2 dataset. Walking sequences are distributed among different folds such that each fold contains each person's one walking sequence. Then iteratively, K models are constructed using ith fold as testing fold in the ith iteration. Each model is trained for 150 epochs with a batch size of 64 and an initial learning rate of 0.001. The learning rate is decreased by the factor of 0.001 based on observed validation loss, resulting in a more stable model. The fitted model is used in the identification phase for score fusion.

Human identification performance evaluation

Table 2 reports the Rank-1 performance of the human identification system using the entire walking sequence with various feature combinations and the proposed LSTM based deep learning model on three benchmark datasets. The LSTM model is created using four different combinations of features. The number of features used from each gait event is given in brackets. The feature combination with body part length and other features produced more than 95% accuracy on all three datasets. Additionally, the GRU-based deep learning model with 105 features performed admirably on all datasets.

In the feature combination, without the body part length feature, the performance is drastically reduced. The LSTM model with 105 features achieved 98.54%, 98.00%, 98.33% accuracy in KGBD, UPCV1, and UPCV2 datasets, respectively. Also, the model trained with 73 features has recorded promising performance in all the datasets. However, for all evaluation metrics, the LSTM model with 105 features consistently recorded the maximum value. The LSTM model with 62 and 30 features has failed to achieve a promising performance. Additionally, it is observed that the GRU model with 105 features performed nearly as well as the LSTM model with 105 features on KGBD but fell short on the other two datasets. Fig. 7 illustrates the proposed system's performance in all K folds of the dataset. Though a small variation in the values, the model showed almost the same performance level on all test folds.

Table 3
Comparison of Rank-1 accuracy of the proposed system for human identification using $Protocol - 1$ with existing methods on KGBD. (Best result is in bold).

Human Identification Methods	Accuracy (%)
Andersson and Araujo [22]	87.7
Yang et al. [10]	95.4
Li et al. [25]	96.56
Khamsemanan et al. [6]	97.5
Liu et al. [19]	97.39
Proposed System (with 105 features + LSTM)	**98.54**

Table 4
Comparison of the proposed system's Rank-1 accuracy using $Protocol - 1$ to existing methods on UPCV1. (Best result is in bold).

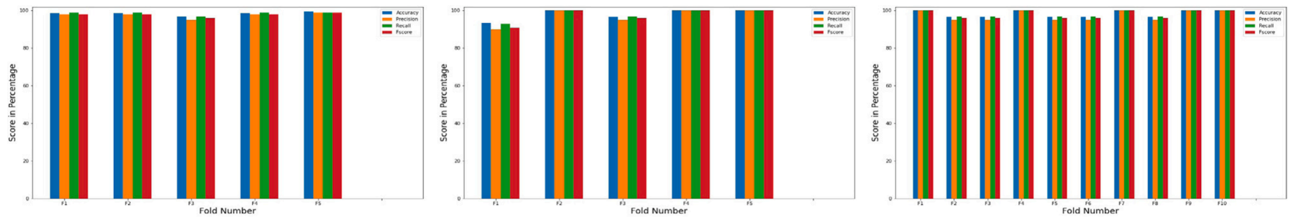
Human Identification Methods	Accuracy (%)
Ince et al. [40]	87.8
Kastaniotis et al. [23]	93.29
Rahman and Gavrilova [41]	93.33
Kastaniotis et al. [4]	95.67
Proposed System (with 105 features + LSTM)	**98.00**

Cumulative Match Characteristic (CMC) curve

In today's smart environment, investigation agencies must consider a group of suspicious individuals before identifying the individual who committed the crime. The CMC test is another evaluation metric for human identification systems that examine the ranking capabilities of the identification system. The proposed system is also evaluated using the CMC test. Fig. 8 shows the CMC curves of the proposed system on three datasets. Consequently, it compares the accuracy of LSTM and GRU-based deep learning models at various Ranks. In the KGBD, the proposed system with LSTM showed 98.53% accuracy at Rank-1 and 99.27% accuracy at Rank-2. In the UPCV1 dataset, the Rank-1 recognition rate is 98.00%, and reached 100% in Rank-3 itself. 98.33% recognition rate at Rank-1 is recorded in the UPCV2 dataset and recognition rate of 99.67% at Rank-3. Overall, the proposed system recorded more than 99% recognition rates within the first three ranks, thus making the proposed system suitable for real-world applications.

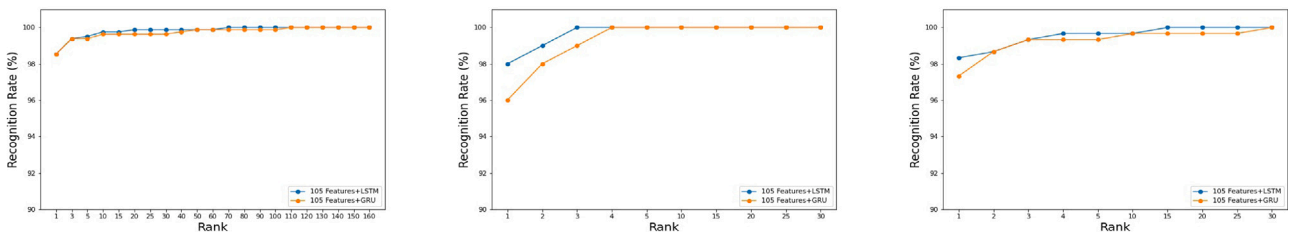
Comparison with existing methods

Tables 3, 4, and 5 report the comparison of Rank-1 accuracy of the proposed work with existing works for human identification on KGBD, UPCV1, and UPCV2, respectively. The results of proposed work compared with five different existing works on the KGBD. It is observed that the proposed system outperformed state-of-the-art works on KGBD in human identification. We achieved 1.15%, 1.04%, 1.98%, 3.14%, and 10.84%, better accuracy than [6,10,19,25], and [22], respectively. The results obtained on UPCV1 is compared with four different existing works using the same evaluation protocol. The proposed work showed a performance improvement of 2.33%, 4.67%, 4.71% and 10.2% as compared to the works of [4,23,41], and [40], respectively on this dataset. Also, proposed system is compared with three existing works on UPCV2 dataset, and achieved 1.28%, 5.92%, and 9.3% better accuracy than [4,24], and [42], respectively on UPCV2 dataset.



(a) (b) (c)

Fig. 7. Rank-1 recognition performance score of proposed system using $Protocol-1$ in K folds: (a) KGBD (b) UPCV1 (c) UPCV2.



(a) (b) (c)

Fig. 8. The CMC curves of the proposed system using $Protocol-1$: (a) KGBD (b) UPCV1 (c) UPCV2.

Table 5
Comparison of the proposed system's Rank-1 accuracy using $Protocol-1$ to existing methods on UPCV2. (Best result is in bold).

Human Identification Methods	Accuracy (%)
Bobillo et al. [42]	89.03
Hosni and Amor [24]	92.41
Kastaniotis et al. [4]	97.05
Proposed System (with 105 features + LSTM)	**98.33**

5.1.2. $Protocol-2$

To determine the proposed system's ability to recognize the gait cycle, we evaluated it based on the evaluation approach of [15]. Since the score fusion subsystem plays no role in evaluating the gait cycle recognition performance, it is bypassed. Again we used the K-fold cross-validation method, and the value of K is 5. Unlike the $Protocol-1$, where all gait cycles extracted from a walking sequence fall into the same fold, in $Protocol-2$, the gait cycles from a walking sequence are randomly distributed among all five folds. Five models are constructed using ith fold for testing in the ith iteration. The training parameters are identical to those used in $Protocol-1$.

Gait cycle recognition performance evaluation

Table 6 reports the Rank-1 gait cycle recognition performance in various experimental set-ups using $Protocol-2$ on KGBD, UPCV1, and UPCV2 datasets. As with $Protocol-1$, the proposed deep learning model using $Protocol-2$ with 105 features performs best in all three datasets. The LSTM based deep learning model with 105 features achieved 97.12%, 96.30%, 99.46% accuracy in KGBD, UPCV1, and UPCV2 datasets, respectively. Also, this model achieved the highest score for Precision, Recall, and F1-Score. We conducted several experiments to evaluate the performance of these 105 features against other machine-learning classification methods, including KNN, RF, SVM, and a two-layer Artificial Neural Network (ANN) model. None of these methods outperformed the LSTM model with 105 feature combinations. Also, the two-layer ANN model achieved significantly less performance than the proposed LSTM model. On the UPCV2 dataset, the deep learning model based on GRU performed similarly to the LSTM model but slightly worse on the KGDB and UPCV1 datasets. Thus, we demonstrated the efficacy of the proposed features and LSTM-based deep learning model in classifying gaits. Also, the LSTM model with all 135 features achieved lesser performance than 105 features on KGBD and UPCV1. But on UPCV2 this performance is almost equal. But it is notable that 135 features will add more complexity to the deep learning model compared to 105 features. Hence it is proved that the proposed LSTM model trained with 105 features exhibits superior performance in recognizing the gait cycles.

Cumulative Match Characteristic Curve

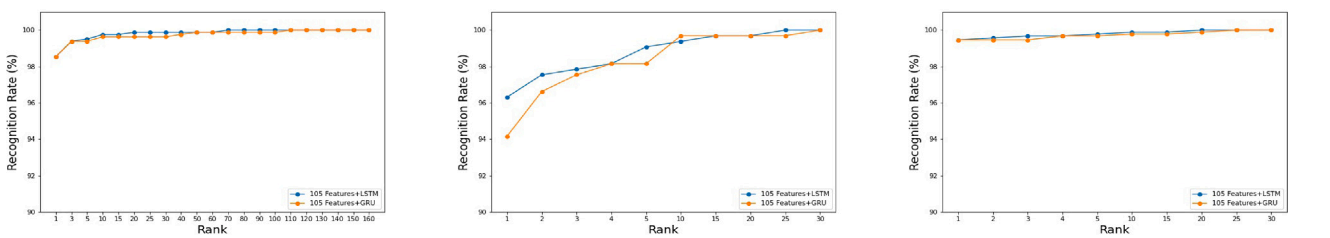
The CMC test curves of the proposed system using the $Protocol-2$ are shown in Fig. 9. Also, it compares the accuracy of the proposed LSTM-based system to that of the GRU-based system at various Ranks. In the KGBD, the proposed system secured a 97.12% Rank-1 recognition rate, Rank-3 recognition rate of 99.3%, and the Rank-10 recognition rate of 99.86%. In UPCV1 dataset, Rank-1 recognition rate is 96.30% and reached 97.85%, 99.38% in Rank-3 and Rank-10, respectively. In UPCV2 dataset, the CMC test reported 99.46%, 99.68% and 99.89% recognition rate in Rank-1, Rank-2, and Rank-3, respectively. Overall, from the CMC test, it is observed that the proposed system with 105 features achieved more than 99% recognition rates in lower level ranks.

Comparison with existing methods

The $Protocol-2$ is a recently introduced evaluation protocol in [15]. The results of the proposed system are compared to those found in [15]. Tables 7 and 8 report the comparison of Rank-1 accuracy on KGBD and UPCV1, respectively. On KGBD though the Rank-1 accuracy is slightly lesser than the existing approach, it reached more than 99% in Rank-2 itself, and in Rank-10, the proposed model achieved 99.86% accuracy. At the same time, [15] reported above 99% and 99.64% recognition accuracy in Rank-3 and Rank-10 respectively, which is slightly lower than ours. Moreover, [15] proposed a deep learning model with 4494592 total parameters (without considering the Softmax layer). In contrast, our proposed system's number of total parameters (without considering the Softmax layer) is just 189210. Thus, our proposed system drastically reduces the computational requirement by 24 times (approximately) as compared to that of [15]. Also, the Rank-1 accuracy obtained by the proposed system on UPCV1 is higher than the existing work. [15] reported that the existing system achieved more than 99% in Rank-7, whereas the proposed system scored more than 99% in Rank-5 itself. So, the proposed system achieved better gait cycle recognition performance with less computational cost.

Table 6
Rank-1 recognition performance of the proposed system with different feature combinations using *Protocol* – 2 (in %) (Best results are in bold).

Experimental Setup	Gait Dataset											
	KGBD				UPCV1				UPCV2			
	Accuracy	Precision	Recall	F1-Score	Accuracy	Precision	Recall	F1-Score	Accuracy	Precision	Recall	F1-Score
BpLF+JDF+ InJDF+JAF (105 Features)+ RF	53.2	59	53.2	53.6	76.6	76	76.6	73.6	94.2	94.6	94.2	93.8
BpLF+JDF+ InJDF+JAF (105 Features)+ KNN (K=1)	64.8	67.6	64.8	64.6	77.4	78.6	77.4	75.2	91.2	92.8	91.2	91.4
BpLF+JDF+ InJDF+JAF (105 Features)+ SVM	84	84.6	83.8	83.8	88.2	88.4	88.2	86.6	97.2	97.8	97.2	97.5
BpLF+JDF+ InJDF+JAF (105 Features)+ 2 Layer ANN	88.25	88.6	88.4	88.4	12.8	6.2	12.8	6.2	90.6	91.2	90.6	90.2
InJDF (30 Features) + LSTM	28.49	27.4	28.4	27	57.33	55.8	57.4	53.6	74.19	76.2	74.4	73.6
InJDF + InJAF (62 Features) + LSTM	49.29	49.2	49.2	49	54.76	54.6	54.6	51.2	82.25	84.8	82.2	82.2
BpLF+JDF+ JAF (73 features) + LSTM	96.39	96.6	96.4	96.4	93.84	95.6	93.8	93.4	99.02	99.4	98.8	98.8
BpLF+JDF+ InJDF+JAF+ InJAF (135 Features) + LSTM	96.78	96.8	96.6	96.6	93.22	95.6	93.2	92.8	99.44	99.8	99.2	99.2
BpLF+JDF+ InJDF+JAF (105 Features) + GRU	96.36	96.8	96.2	96.2	94.14	95.4	94	93.6	**99.46**	**99.8**	**99.2**	**99.2**
BpLF+JDF+ InJDF+JAF (105 Features) + LSTM (Proposed System)	**97.12**	**97**	**97**	**97**	**96.30**	**97.2**	**96.2**	**96**	**99.46**	**99.8**	**99.2**	**99.2**



(a) (b) (c)

Fig. 9. The CMC curves of the proposed system using *Protocol* – 2: (a) KGBD (b) UPCV1 (c) UPCV2.

Table 7
Comparison of the proposed system's Rank-1 recognition accuracy using *Protocol* – 2 to existing methods on KGDB. (Best result is in bold).

Gait Cycle Recognition Methods	Accuracy (%)
Bari and Gavrilova [15]	**98.08**
Proposed System (with 105 features + LSTM)	97.12

Table 8
Comparison of the proposed system's Rank-1 recognition accuracy using *Protocol* – 2 to existing methods on UPCV1. (Best result is in bold).

Gait Cycle Recognition Methods	Accuracy (%)
Bari and Gavrilova [15]	95.30
Proposed System (with 105 features + LSTM)	**96.30**

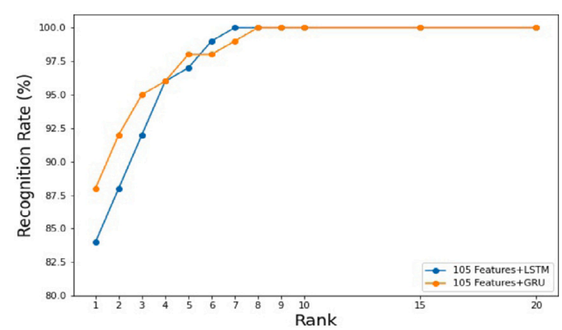


Fig. 10. CMC test curve obtained in *Random Split* method on KS20.

5.2. Experiments on multi-view dataset

We conducted several experiments on the multi-view skeleton-based gait dataset KS20 to demonstrate the proposed system's effectiveness in view-invariant scenarios. The samples in the dataset provide a single gait cycle. So the data augmentation is performed to increase the number of gait cycles as follows. As already explained, consecutive three peaks in ankle distances make a gait cycle. So initially, an additional gait cycle starting from another leg is constructed by placing the set of frames between the first and second peaks immediately after the third peak. Further, the number of gait cycles is increased by rotating all skeleton joints in both gait cycles about different axes after translation and normalization steps mentioned in Section 3.1. Then the performance is evaluated based on the following two existing evaluation protocols [43]:

- *Random Split*: Randomly selected two samples from each viewpoint are used for training, and the remaining one sample from each viewpoint is used for testing.
- *Cross-view Split*: Test on all samples from each viewpoint while considering samples from remaining viewpoints for training.

As both evaluation protocols aim to recognize the gait cycles as in *Protocol* – 2 used for single-view dataset, we excluded the score fusion part during experiments. Also, the augmented gait cycles are used only during training. To better analyze the proposed system, we performed a number of experiments on a multiview dataset using the same set of training parameters used for single-view datasets. The Rank-1 results obtained with various feature combinations on *Random Split* protocol are shown in Table 9. The experiment with the combination of 105 features and LSTM stood first in identifying persons using the gait. The comparison of the proposed system for view-invariant gait recognition with existing works using *Random Split* is given in Table 10. The proposed system outperformed most of the existing approaches.

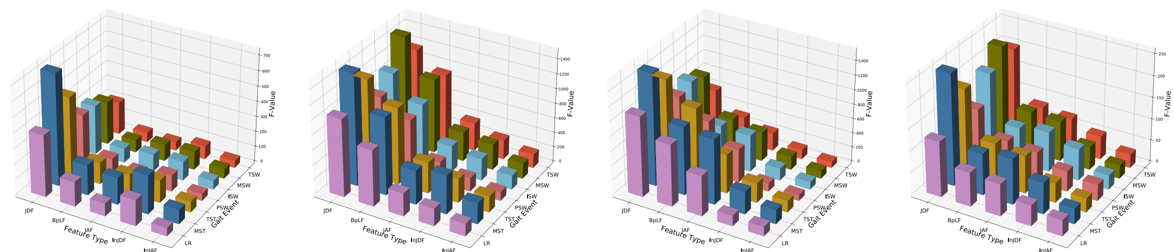
The CMC curve obtained using *Random Split* experimentation is shown in Fig. 10. The Rank-1, Rank-5, and Rank-10 accuracy obtained using *Cross-view Split* up using a combination of 105 features and the proposed LSTM model along with the comparison of Rank-1 accuracy with existing methods on all views are given in Table 11. The proposed system achieved better recognition accuracies compared to existing methods in two different views. The results of [44] are based on [43].

5.3. Ablation experiments

Additionally, as reported in [46], we conducted ablation experiments to ascertain the relevance of layers in the proposed deep learning model in identifying gait cycles. These experiments evaluate the models on protocols that do not require score fusion, i.e., *Protocol* – 2 of single-view datasets and *Random Split* of the multi-view dataset. The findings achieved by eliminating the proposed deep learning model's various layers on three datasets are shown in Table 12. The results indicate that

Table 9
Rank-1 recognition performance of the proposed system on KS20 dataset using *Random Split* protocol (in %) (Best result is in bold).

Experimental Setup	Accuracy	Precision	Recall	F1-Score
BpLF+JDF+ InJDF+JAF (105 Features)+ KNN (K=1)	47	52	47	47
BpLF+JDF+ InJDF+JAF (105 Features)+ RF	48	50	48	45
BpLF+JDF+ InJDF+JAF (105 Features)+ SVM	65	68	65	64
BpLF+JDF+ InJDF+JAF (105 Features)+ 2 Layer ANN	12	2	12	3
InJDF (30 Features)+LSTM	31	29	31	29
InJDF + InJAF (62 Features)+LSTM	36.25	37	36	35
BpLF+JDF+ JAF (73 features)+LSTM	86	87	86	86
BpLF+JDF+ InJDF+JAF+ InJAF (135 Features)+LSTM	86	88	86	86
BpLF+JDF+ InJDF+JAF (105 Features)+ GRU	84	87	84	84
BpLF+JDF+ InJDF+JAF (105 Features)+LSTM (Proposed System)	**88**	**89**	**88**	**88**



(a) (b) (c) (d)

Fig. 11. Comparison of total F-Value of feature groups in gait events: (a) UPCV1 (b) KGBD (c) UPCV2 (d) KS20 (130°).

Table 10
Comparison of Rank-1 accuracy with existing methods on KS20 dataset using *Random Split* protocol. (Best result is in bold).

Human Identification Method	Accuracy (%)
Pose Gait [44]	70.5
Context Unware [37]	79.33
3D spatial–temporal [45]	87.63
Context Aware [37]	88.67
Self-Supervised [43]	**92**
Proposed System (with 105 Features and LSTM Model)	88

Table 11
Accuracy (in %) of proposed system on KS20 dataset using *Cross-view Split* protocol and comparison with existing methods.

View	Pose Gait [44]	Self-Supervised [43]	Proposed System		
	Rank-1	Rank-1	Rank-1	Rank-5	Rank-10
0°	24.6	**48.8**	25	46.67	76.67
30°	19.1	**53.6**	45	75	91.67
90°	29.7	54.9	**73.33**	93.33	98.33
130°	27.3	44.5	**78.33**	95	100
180°	25	**57.5**	36.67	78.33	90

omitting any layer from the proposed model has a noticeable impact on performance. Thus, each layer contributes to the precise recognition of gait cycles.

5.4. Further empirical analysis (ANOVA test)

To investigate the features selected and the methodology used, we performed the empirical analysis based on [47], on features extracted and results obtained on different folds using different combination of features and the proposed LSTM based deep learning model.

5.4.1. Feature set analysis

We used a three-step procedure to analyze the extracted features. The theory of ANOVA states that the greater F-value indicates high discriminative capability [47]. So, initially, using the ANOVA test, the F-value of each 135 features is determined. On all datasets, the BpLF features secured the highest F-value, whereas InJAF features had lower F-values. Then, as the features extracted in the proposed system are grouped under five categories as explained in Section 3.3, the F-values of features from each group are summed up. Finally, the feature group F-values are sorted to select the discriminative feature types. ANOVA test showed that on all the datasets, InJAF features are the least discriminative. The same is reflected in the experimental results also. The comparison between the total F-values of feature groups in each gait event on different datasets is shown in Fig. 11.

5.4.2. Analysis of statistical significance of proposed human identification system

The statistical significance of all the different experimental settings with various classifiers is tested by conducting One-way analysis of variance test on the results obtained by various experimental settings based on $Protocol - 2$ from K-folds on KGBD, UPCV1, and UPCV2 as it is done in [47]. The P-value obtained from the One-way analysis of variance test is 1.227e–25, 9.2523e–27, 4.967e–53 on UPCV1, UPCV2, and KGBD, respectively. Similarly, we conducted these tests on results obtained on the KS20 dataset based on Cross-view split-up and obtained 0.0034 P-value from the One-way analysis of variance. The null hypothesis H_0 considered is 'all classifiers will result in same average recognition accuracies'. From the analysis, it is clear that $P-Value < \alpha$ on all the datasets, where $\alpha = 0.05$. Hence H_0 is rejected. The tests are conducted following the procedures in [48].

5.5. Computational cost analysis and discussion

In the proposed system, the total computational cost depends on three subsystems, namely: feature extraction, score fusion, and LSTM based deep learning model. The proposed system extracted features from gait events considering single and consecutive two frames. The estimated time complexity of feature extraction is $O(n)$, where n is the number of frames in the walking sequence. The estimated time complexity of score fusion is $O(L)$, where L is the number of labels in the dataset.

Table 12
Ablation experiments results.

Excluded Layers	Dataset			
	KGDB	UPCV1	UPCV2	KS20
	Accuracy (%)	Accuracy (%)	Accuracy (%)	Accuracy (%)
Second LSTM	94.14	92	97	75
First BN	92.96	95.45	99	84
Second BN	94.86	93.85	98.82	75
First Dense	95.95	93	98.65	77

The total execution time for a single epoch of the deep learning model is given in Eq. (9) as suggested in [49].

$$E = pT_b, \tag{9}$$

Where, p is the number of batches, and T_b is the time required for forward and backward passes on a single batch and computed as defined by Eq. (10).

$$T_b = \sum_{i=0}^{l} b_{M(i)}, \tag{10}$$

Where, l is the number of layers, $b_{M(i)}$ is the batch execution time estimate for layer i, $M(i)$ is of type of layer i and it depends on layer type.

The proposed deep learning model has two LSTM, two Batch Normalization, one Dense and Softmax layers (Dense). The LSTM is local in time and space [18]. So, the input length does not affect the network's storage requirements [50]. Therefore for each timestamp, the time complexity per weight is $O(1)$ and total time complexity per timestamp is $O(W_L)$, where, W_L is the number of weights that is approximately equal to total parameters in the LSTM layer. The Batch Normalization mainly involves finding minibatch mean and variance. The estimated time complexity of each of these per batch is $O(b)$, where, b is the batch input length. The number of computations in a Dense layer also proportional to the number of weights in a Dense layer, and it is approximately $O(W_d)$, where, W_d is the number of weights in the Dense layer. Therefore using Eqs. (9) and (10), the total time required for forward and backward passes on a single epoch is estimated as defined by Eq. (11).

$$Est = p * (2 * O(W_L) + 2 * O(b) + 2 * O(W_d)), \tag{11}$$

The $Protocol - 1$ has all three subsystems of the proposed system. So the total estimated time for e epochs is given in Eq. (12).

$$Est_{pt1} = O(n) + (p * (2 * O(W_L) + 2 * O(b) + 2 * O(W_d))) * e + O(L), \tag{12}$$

The $Protocol - 2$, *Random*, and *Cross-view Splits* evaluation protocols have first two subsystems of the proposed system. So the total estimated time for e epochs is given in Eq. (13).

$$Est_{pt1} = O(n) + (p * (2 * O(W_L) + 2 * O(b) + 2 * O(W_d))) * e, \tag{13}$$

In all the evaluation protocols, the number of features and the deep learning model parameters significantly affect the total computations required. The number of parameters in the proposed LSTM based deep learning model is very less when compared to other approaches. The deep learning model suggested in [15] has 4494592 parameters, whereas the number of parameters in our proposed system is just 189210; thus reducing the computational cost by 24 times approximately.

All the experiments are conducted on a system with configuration: Intel Core i7-10750H CPU @ 2.60 GHz, 2592Mhz, 6 Core, 12 Logical Processors, 16 GB RAM, and GPU of NVIDIA GeForce GTX 1650.

5.6. Limitations

The proposed approach computes the mean and standard deviation of the various inter and intra-frame distances and angles between the joints as the feature vector. This effectively eliminates the effect of incorrect angles or distances derived from frames with occluded or noisy joint data. Simultaneously, the summarization process may suppress the benefits of the correct frames and may have a minor effect on performance.

6. Conclusion

This paper proposes a new gait event level feature set and a deep learning model based on LSTMs for human identification using 3D skeleton-based gait features. For each gait event, a summary of various intra and inter-frame angles and distances is derived, resulting in a significant reduction in the number of features and, as a result, the number of computations. It is also an attempt to reduce the impact of noise and occluded body parts on distance and angles. The proposed features are tested with a variety of classifiers, including SVM, KNN, Random Forest, ANN, and a GRU-based deep learning model. A series of experimental setups with various feature combinations are used to determine the parameters of the proposed LSTM-based deep learning model. The ablation experiments have been carried out to demonstrate the contributions of various layers.

Proposed system's performance is evaluated using various benchmark protocols on three publicly available single-view and on a multi-view gait datasets. The experimental results showed that the proposed approach achieved greater than 96% accuracy on all three single-view datasets on all the protocols. During the CMC test, the proposed system also achieved a recognition rate of over 99% in low-level ranks. Furthermore, experiments on a multiview gait dataset revealed that low-level ranks were more than 96% accurate, making it suitable for practical applications. Also, the statistical analysis test performed on the features demonstrated the extracted features' discriminating ability in classification. We performed a One-way ANOVA test on the results obtained from the K-fold cross-validation process to test the statistical significance between the different individual experimental setups. The P-Value result indicates that the proposed approach is significantly different from others.

In the future, we will focus on extracting a large number of more efficient gait event features and more robust deep learning models based on [51–53] due to their capability of capturing long-range dependability between data and superior performance in attaining their target objectives.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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